

Malaria Detection: Report

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Google Collab Link: https://colab.research.google.com/drive/1sFd3C-V1FjZfiX_CQq-O-sX6PsWqEwnF?usp=sharing

This project developed a Convolutional Neural Network (CNN) to detect malaria parasites in blood cell images, addressing significant dataset challenges including mixed image formats and severe class imbalance.

I. Project Overview

- Training Set: 4,800 images (4,000 negative, 800 positive)
- Validation Set: 2,566 images (1,531 negative, 1,035 positive)
- Class Imbalance Ratio: 5:1 (training), 1.5:1 (validation)
- Image Formats: Mixed TIFF and PNG files
- Dimensions: Variable, ranging from 30×32 to 67×86 pixels

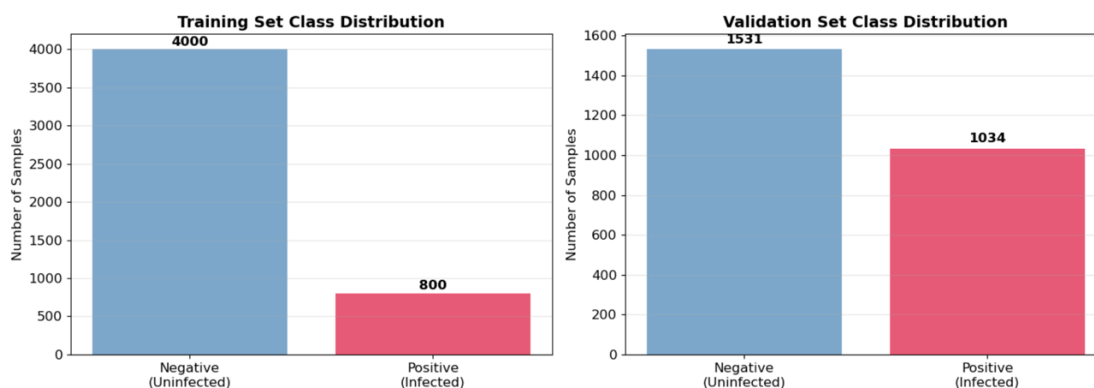
II. Challenges Identified

Challenge 1: Mixed Image Formats

The dataset contained images in both TIFF and PNG formats, with all negative training samples in TIFF (4,000 images) and most positive samples in PNG (800 training + all validation). This created a critical risk: the model could learn to distinguish file format artifacts rather than actual malaria parasites.

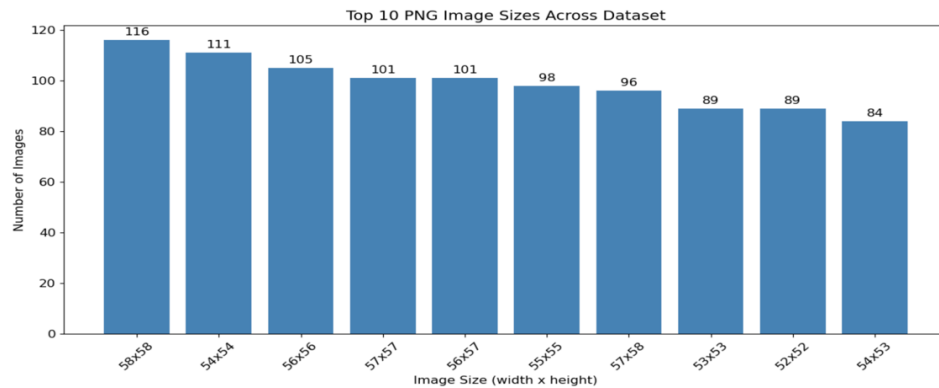
Challenge 2: Severe Class Imbalance

With a 5:1 ratio of negative to positive samples in the training set, the model could achieve 83% accuracy by simply predicting "negative" for every image, completely ignoring the minority class.



Challenge 3: Inconsistent Image Dimensions

Images varied significantly in size (30×32 to 67×86 pixels), making it impossible to batch process them or form uniform tensors for neural network input without standardization.



Challenge 4: Overfitting Tendency

Initial CNN architectures with large dense layers (512 and 256 neurons) achieved 100% training accuracy but experienced exploding validation loss, indicating severe memorization rather than generalization.

III. Preprocessing Strategy

Step 1: Format Standardization

Decision: Convert entire dataset to PNG format.

Rationale:

- Over 45% of the dataset was already in PNG
- PNG is a widely accepted, lossless format in biomedical imaging
- Standard malaria image benchmarks (NIH Malaria Dataset with 27,558 images) use PNG/JPEG
- Ensures compatibility with pretrained models (ResNet, VGG) that expect standard RGB inputs
- Eliminates file format as a confounding variable

Implementation:

- Converted all TIFF images to PNG
- Removed system files (Thumbs.db) found in validation directories

Step 2: Dimension Standardization to 64×64

The 64×64 resolution was chosen based on mathematical and biological considerations:

- Dataset dimensions cluster around 55 pixels (range: 30-67 pixels)
- Standard 224×224 would require up to 700% upscaling for smallest images, introducing severe interpolation artifacts
- 64×64 is the closest power-of-two resolution matching the dataset's natural scale
- Larger images downsample cleanly with minimal information loss
- Smaller images undergo only mild upscaling without destructive interpolation

Drawbacks of Cropping or Direct Resizing:

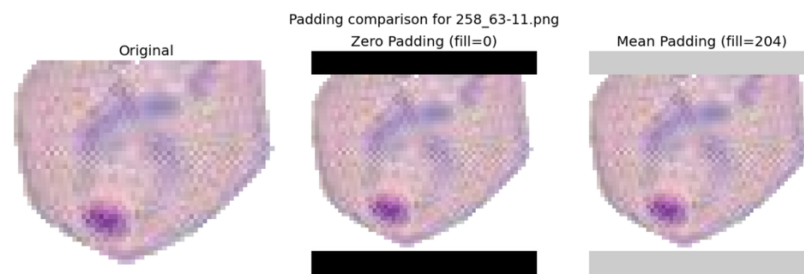
- **Cropping:** Risk of removing parasites at cell edges
- **Direct Resizing:** Distorts aspect ratio, altering critical cell morphology

Solution: Downsample-Then-Pad Approach

1. **Downsample first:** Reduce resolution while maintaining spatial relationships and morphology
2. **Pad second:** Add borders to reach 64×64 without stretching or compressing cells

Step 3: Artifact-Free Padding

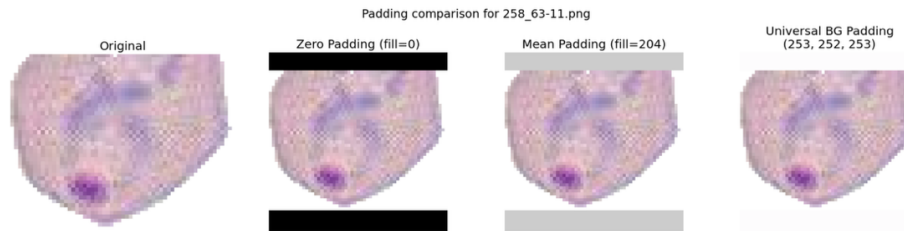
- **Zero-padding (black):** Created sharp, high-contrast borders unnatural for microscope images
- **Mean-padding:** Produced visible gray bands as dark parasite pixels pulled the average downward



Solution: Universal Background Color

Analyzed 5×5-pixel blocks from the four corners of all training images to determine the true microscope slide background color. Corners consistently contain empty background, providing an accurate estimate.

Result: Universal background color of (253, 252, 253) RGB



Benefits:

- Seamless visual integration with microscope background
- No artificial edges or contrast artifacts
- Model focuses solely on biological features rather than padding borders

Step 4: Data Normalization

Process: Divided all pixel values by 255 to normalize from [0, 255] to [0, 1] range.

Benefits:

- Neural networks initialize weights with small random numbers
- Large integer inputs (0-255) create massive mathematical variance
- Destabilizes gradients during backpropagation
- Normalized [0, 1] range creates smooth error landscape
- Allows optimizer to converge significantly faster

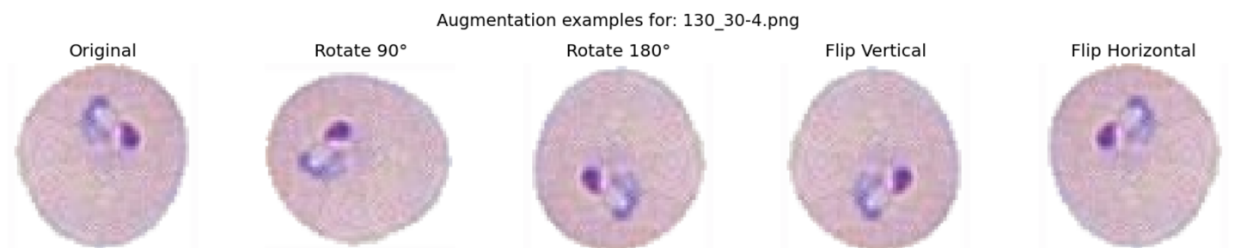
IV. Class Imbalance Mitigation(Data Augmentation)

Target: Achieve perfect class parity (4,000:4,000)

Calculation: Expansion factor of 5 required ($4000 / 800 = 5$)

Implementation: For each positive image, generate 4 synthetic copies using:

1. **Rotate 90°**
2. **Rotate 180°**
3. **Vertical Flip**
4. **Horizontal Flip**



Why These Transformations?

- **Mathematically lossless:** Rigid geometric operations preserve exact pixel values
- **Biologically safe:** Blood cells are rotation-invariant; parasites maintain diagnostic features
- **No interpolation artifacts:** Unlike random rotations or blurring
- **Preserves color integrity:** Critical for parasite identification

Result: Balanced training set of 8,000 images (4,000 negative, 4,000 positive)

V. Model Architecture Evolution

Configuration:

- Multiple convolutional blocks
- Dense layers: 512 and 256 neurons
- No Batch Normalization
- No L2 Regularization

Problems:

- Training accuracy: 100%
- Validation loss: Exploding
- Model memorized training data rather than learning generalizable features

Key Modifications:

1. **Drastically Reduced Dense Layer Size**
 - Changed from 512/256 to 64 neurons
 - Rationale: Malaria parasites are simple shapes (blobs/rings); excessive parameters enable memorization
2. **Added Batch Normalization**
 - Applied after every convolutional layer and before activation
 - Stabilizes training by normalizing activations
 - Has inherent regularizing effect
 - **Non-trainable Parameters:** 576 (288 channels $\times$ 2 statistics per channel)
 - Tracks moving mean and variance for each channel
 - 32 (Conv1) + 64 (Conv2) + 128 (Conv3) + 64 (Dense) = 288 channels
3. **Added L2 Regularization**
 - Penalizes large weights
 - Forces model to be "humble" rather than arrogantly predicting extreme probabilities (0.9999)
 - Encourages smoother, more generalizable predictions

Final Architecture:

Conv Block 1: 32 filters → BatchNorm → MaxPooling → Dropout

Conv Block 2: 64 filters → BatchNorm → MaxPooling → Dropout

Conv Block 3: 128 filters → BatchNorm → MaxPooling → Dropout

Flatten

Dense: 64 neurons → BatchNorm → Dropout

Output: 1 neuron (sigmoid activation)

VI. Training Configuration

Training Strategy

- **Loss Function:** Binary crossentropy
The standard loss function used for binary classification problems (two mutually exclusive classes, e.g., "Parasitized" vs. "Uninfected"). Our model uses a Sigmoid activation in the final layer, which outputs a probability value between 0 and 1. Binary Crossentropy is mathematically designed to work with Sigmoid outputs. It heavily penalizes the model for confident but wrong predictions (e.g., predicting 0.99 probability for a negative class image)
- **Optimizer:** Adam (adaptive learning rate)
- **Callbacks:**
 - Early stopping (monitors validation loss)
 - Learning rate reduction on plateau
- **Metrics Tracked:** Accuracy, Precision, Recall, Loss (both training and validation)

Regularization Techniques Used

Overfitting occurs when the model memorizes the training data but fails to generalize to new images. The following techniques were implemented to prevent this.

1. **Batch Normalization:** A technique that normalizes the inputs of each layer, so they have a mean of 0 and a standard deviation of 1. It prevents "internal covariate shift," meaning the later layers don't have to constantly readjust to shifting data distributions from earlier layers. It allows the model to train faster and accept higher learning rates.
2. **Dropout:** A technique where randomly selected neurons are ignored (set to zero) during training. We used rates of 0.2, 0.3, and 0.5. It forces the network to learn robust features rather than relying on specific pathways. It effectively simulates training many different smaller neural networks simultaneously.
3. **L2 Regularization (Weight Decay):** A penalty added to the loss function proportional to the square of the magnitude of the weights (regularizers.l2(0.001)). It forces the model weights to remain small. Large weights are often a sign that the model is overfitting to

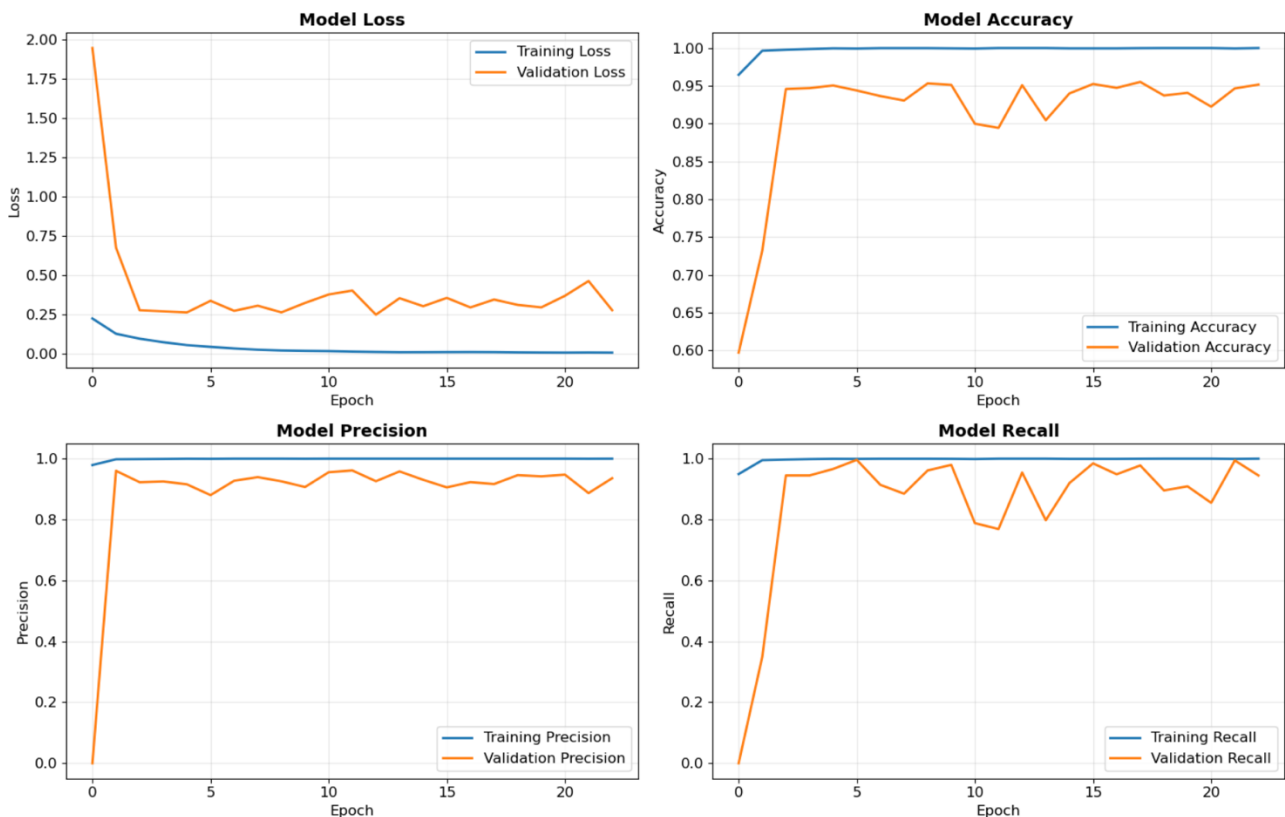
noise in the training data. By keeping weights small, the model becomes smoother and more general.

Training Callbacks Used

Callbacks are automated tools that monitor the training process and intervene based on performance metrics.

1. **Early Stopping:** A mechanism that monitors the validation loss (val_loss) and stops training if it does not improve for a set number of epochs (Patience = 10). Prevents wasting computational resources on a model that has stopped learning. We use `restore_best_weights=True` to ensure that even if the model overfits in later epochs, we revert to the version that performed best on the validation data.
2. **Reduce ROOnPlateau (learning Rate Deacy):** An algorithm that reduces the learning rate by a factor of 0.5 if the validation loss stops improving for 5 epochs. In the beginning, a higher learning rate allows the model to learn quickly. As the model approaches the optimal solution, a smaller learning rate is needed to make tiny adjustments and settle into the "global minimum" without overshooting.

Training History - Improved CNN Model (BatchNorm + L2 Regularization)



VII. Results

Analysis of Model Performance

In the context of medical diagnostics like malaria detection, **Recall (Sensitivity)** is the most critical metric because the cost of a "False Negative" is extremely high missing an infected cell means a patient goes undiagnosed and untreated.

Validation Metrics:

- **Accuracy:** ~95%
- **Training Stability:** Smooth convergence without catastrophic failures
- **Generalization:** No significant overfitting; training and validation curves aligned

Classification Report:				
	precision	recall	f1-score	support
Negative	0.97	0.95	0.96	1531
Positive	0.93	0.95	0.94	1034
accuracy			0.95	2565
macro avg	0.95	0.95	0.95	2565
weighted avg	0.95	0.95	0.95	2565

Detailed Metrics:

Accuracy: 0.9509

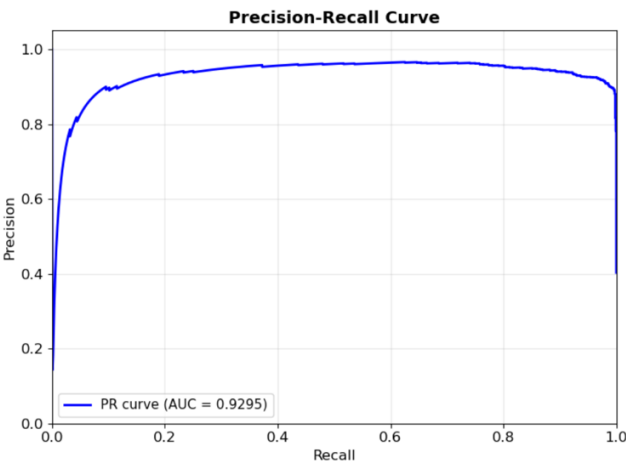
Precision: 0.9259

Recall: 0.9545

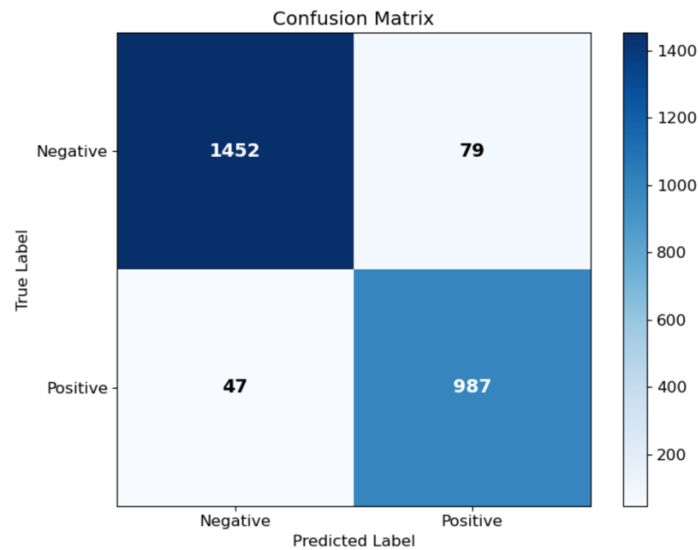
F1-Score: 0.9400

ROC-AUC: 0.9810

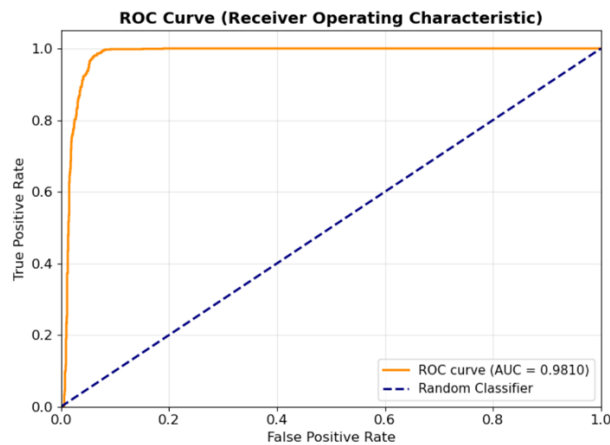
This classification report demonstrates the model's consistent performance across varying metrics, specifically highlighting the support, F1-score, and averaging methods. The **support** column represents the number of actual occurrences of each class in the dataset, indicating a test set comprised of 1,531 negative samples and 1,034 positive samples. The **f1-score**, which acts as the harmonic mean of precision and recall, remains high at 0.96 for the negative class and 0.94 for the positive class, signifying that the model maintains a reliable balance between sensitivity and correctness. Additionally, the **macro avg** (0.95) provides an unweighted mean that treats both classes as equally important regardless of their size, while the **weighted avg** (0.95) accounts for the support of each class, confirming that the model's high performance is sustained even when adjusting for the slight imbalance between the number of negative and positive samples.



The Precision-Recall curve evaluates the trade-off between the model's ability to detect all positive cases (Recall) and the accuracy of its positive predictions (Precision), effectively highlighting performance on the minority class (infected cells). An AUC of 0.9295 confirms that the model maintains high precision even at high sensitivity levels, ensuring that the necessary focus on minimizing false negatives in a medical context does not result in an excessive number of false alarms.



This confusion matrix illustrates the performance of the classification model used for detecting malaria infections, showcasing a strong ability to distinguish between uninfected and parasitized samples. The results highlight a high number of True Positives (987) compared to a relatively low number of False Negatives (47). In the context of medical diagnostics, minimizing False Negatives is paramount to achieving high recall (sensitivity), as failing to detect an active malaria infection could lead to dangerous delays in treatment. The low count of False Negatives demonstrates that the model prioritizes patient safety by effectively capturing most positive infection cases.



The Receiver Operating Characteristic (ROC) curve illustrates the diagnostic ability of the binary classifier by plotting the True Positive Rate (Sensitivity) against the False Positive Rate (1 - Specificity) at varying threshold settings. The resulting Area Under the Curve (AUC) of 0.9810 indicates exceptional discriminatory performance, demonstrating that the model can distinguish between infected and uninfected cells with near-perfect reliability independent of the specific decision threshold used.

Key Achievements

- Eliminated Format Bias:** Model learns biological features, not file artifacts
- Balanced Classes:** Data augmentation achieved 1:1 ratio without synthetic noise

3. **Preserved Image Quality:** Artifact-free preprocessing maintained diagnostic features
4. **Stable Training:** Optimized architecture and batch size prevented overfitting
5. **Practical Dimension:** 64×64 resolution balances detail preservation with computational efficiency

VIII. Conclusions

1. **Preprocessing is Critical:** More than half the project effort focused on proper data preparation; this foundation enabled model success
2. **Domain Knowledge Matters:** Understanding microscope imaging and parasite morphology informed decisions about padding, augmentation, and resolution
3. **Simpler is Better:** Reducing model complexity (64 vs 512 neurons) improved generalization for simple biological shapes
4. **Batch Normalization is Essential:** Stabilizes training and provides regularization without explicit penalties
5. **Batch Size Impacts Stability:** Smaller batches navigate difficult samples more gracefully than large batches

This approach demonstrates that careful preprocessing and thoughtful architectural choices can achieve high accuracy on medical imaging tasks without requiring:

- Massive datasets
- Pretrained models
- Expensive computational resources